

AI Data Insights Engine

LLM-Powered Automated Analytics for Business Teams

Claude API · Python · Streamlit · Pandas · Plotly

Rahul Sangral

Data Analyst · AI Developer · Founder, Clavix Technologies

rahul.rishusangral@gmail.com · India · github.com/sangralrahul

Table of Contents

- 01. Executive Summary
- 02. Problem Statement & Market Context
- 03. Solution Overview
- 04. System Architecture
- 05. Technology Stack Deep Dive
- 06. Core Features & Modules
- 07. LLM Integration Strategy (Claude API)
- 08. Data Ingestion Pipeline
- 09. Visualisation Engine
- 10. Anomaly Detection Module
- 11. Security & Data Privacy
- 12. Performance Benchmarks
- 13. User Experience Design
- 14. Deployment & Infrastructure
- 15. Testing & Quality Assurance
- 16. Known Limitations & Mitigations
- 17. Roadmap & Future Enhancements
- 18. Business Impact & ROI
- 19. Conclusion
- 20. References & Appendix

01. Executive Summary

AI Data Insights Engine is an open-source, LLM-powered analytics platform that transforms raw CSV and Excel datasets into actionable business intelligence within seconds. Designed for analysts, product managers, and non-technical stakeholders, it eliminates the need for manual data wrangling and report generation.

Built on the Claude API (Anthropic), Python, and Streamlit, the tool auto-generates natural-language summaries, interactive visualisations, and anomaly alerts — reducing manual analysis time by an average of 60% across tested workloads.

Key Outcomes

- 60% reduction in average analysis turnaround time
- Zero-code interface — no SQL or Python skills required for end users
- Supports datasets up to 500 MB with automatic chunking
- Integrates with Claude 3 Haiku / Sonnet for cost-optimised inference
- Fully open-source under MIT licence

02. Problem Statement & Market Context

Business teams across industries generate vast quantities of structured data — sales records, operational logs, customer interactions, financial ledgers — yet deriving insights from this data requires skilled data analysts and significant time investment. According to IDC (2024), the average analyst spends 42% of their time cleaning and formatting data before any analysis begins.

Core Pain Points Identified

- Manual CSV inspection is slow and error-prone
- Non-technical stakeholders cannot independently query or interpret datasets
- Existing BI tools (Tableau, Power BI) require significant setup and licensing costs
- Ad-hoc reporting bottlenecks analyst capacity — teams report 15–20 report requests per week
- Anomalies go undetected until they escalate into business-critical issues

The AI Data Insights Engine addresses these pain points by placing LLM reasoning directly in the analytics workflow, allowing any team member to upload a dataset and receive a plain-English summary of what the data means, what has changed, and what requires attention.

03. Solution Overview

The platform operates as a single-page Streamlit application. Users upload a CSV or Excel file through a drag-and-drop interface, and the engine orchestrates a three-stage pipeline:

Stage 1 — Ingestion & Profiling

Pandas and custom preprocessing modules parse the file, infer data types, detect missing values, calculate descriptive statistics, and generate a structured data profile JSON object.

Stage 2 — LLM Analysis

The data profile is passed to the Claude API alongside a carefully engineered prompt template. Claude returns a structured JSON response containing: an executive summary, key trend observations, anomaly flags, column-level insights, and recommended follow-up questions.

Stage 3 — Visualisation & Export

Plotly generates interactive charts (time series, bar, scatter, heatmap) based on Claude's recommendations. The full report is exportable as PDF or HTML.

04. System Architecture

The application follows a modular single-service architecture optimised for local and cloud deployment.

Component Diagram (Textual)

- [Browser UI] !' Streamlit Frontend (Port 8501)
- Streamlit Frontend !' [File Upload Handler] !' [Data Profiler (Pandas)]
- Data Profiler !' [Prompt Builder] !' [Claude API via HTTP]
- Claude API !' [Response Parser] !' [Chart Generator (Plotly)]
- Chart Generator !' [Report Builder] !' [PDF/HTML Export]
- All data processed in-memory — no persistent storage of user data

Design Principles

- Stateless: no user data persisted between sessions
- Modular: each stage is independently testable
- Cost-aware: automatic model selection (Haiku vs Sonnet) based on dataset size
- Offline-ready: core profiling runs without API access; LLM step is optional

05. Technology Stack Deep Dive

Python 3.11

Core runtime. Chosen for its mature data science ecosystem, type hints for maintainability, and excellent async support.

Streamlit 1.32

Provides the interactive web UI with zero frontend code. Session state management handles multi-step workflows. Custom CSS injected for dark-theme branding.

Pandas 2.1 & NumPy 1.26

Data ingestion, type inference, statistical profiling, and preprocessing. Optimised with chunked reading for large files.

Claude API (Anthropic)

Used for natural-language generation of insights. Prompt templates are versioned and stored in YAML configuration files for easy iteration. Supports claude-3-haiku-20240307 and claude-3-sonnet-20240229.

Plotly 5.18

Interactive charting library. Charts are embedded as Streamlit components and exported as static SVG/PNG for PDF reports.

06. Core Features & Modules

Auto-Profiler

- Detects column types: numeric, categorical, datetime, boolean, free text
- Reports: null %, cardinality, min/max/mean/std, top-N categories
- Flags: mixed-type columns, duplicate rows, outlier-dense columns

Smart Summary Generator

- Generates 3-paragraph executive summary in plain English
- Adapts language based on detected domain (finance, healthcare, retail, etc.)
- Highlights strongest correlations and significant distributions

Trend Detector

- Identifies monotonic trends in time-series columns
- Calculates period-over-period % change
- Surfaces top-3 positive and negative trend signals

Anomaly Alert Engine

- Z-score method for numeric outliers (threshold configurable)
- IQR method as secondary pass for skewed distributions
- Isolation Forest integration (optional, scikit-learn) for multivariate anomalies

07. LLM Integration Strategy

Effective LLM integration requires careful prompt engineering, token management, and fallback handling. The system uses a layered prompting approach:

Prompt Architecture

- System prompt: defines Claude's role as a senior data analyst with domain awareness
- Context block: injects dataset schema, statistics, and sample rows (max 20 rows)
- Task prompt: specifies required output format as structured JSON
- Constraints: instructs Claude to cite specific column names and numeric values

Token Optimisation

- Large datasets are summarised before being sent (statistical aggregates, not raw rows)
- Context window: 4K tokens for Haiku, 20K for Sonnet — auto-selected based on dataset complexity
- Output is streamed for responsive UX on longer analyses

Fallback Handling

If the Claude API is unavailable, the tool falls back to a rule-based summary generator that produces a structured (though less nuanced) plain-text report from the Pandas profile alone.

08. Data Ingestion Pipeline

The ingestion module is built for robustness across real-world dirty datasets.

- Supports: CSV (UTF-8, Latin-1, CP1252), XLSX, XLS, ODS
- Automatic encoding detection via chardet
- Chunked reading: 50K rows per chunk for files > 100 MB
- Schema inference: tries 5 datetime formats before falling back to string
- Deduplication: flags duplicate rows, optionally removes them
- Column normalisation: strips whitespace, lowercases headers, replaces spaces with underscores

09. Visualisation Engine

Charts are generated based on Claude's recommendations and column type heuristics. The engine selects chart types automatically:

- Numeric + datetime !' line chart (trend over time)
- Categorical with < 20 unique values !' bar chart
- Two numeric columns !' scatter plot with regression line
- Correlation matrix for datasets with > 5 numeric columns
- Distribution histogram for skewed numeric columns
- Heatmap for time-day aggregations (e.g., hourly sales by weekday)

All charts use the custom dark purple theme matching the application branding. Annotations are added for anomalies and trend inflection points.

10. Anomaly Detection Module

Anomaly detection runs as a parallel async task during LLM analysis to avoid blocking the UI.

Detection Methods

- Z-Score (primary): flags values > 3 standard deviations from mean
- IQR (secondary): catches outliers in non-normal distributions
- Isolation Forest (optional): detects multivariate anomalies using scikit-learn
- Rule-based: detects impossible values (e.g., negative age, future dates in historical data)

Alert Format

Each anomaly alert includes: column name, row index, detected value, expected range, severity level (Low / Medium / High), and a plain-English explanation generated by Claude.

11. Security & Data Privacy

Data privacy is a first-class concern, particularly for enterprise users who may upload sensitive datasets.

- No data persistence: all processing is in-memory; files are not written to disk beyond the session temp directory
- Session isolation: each Streamlit session has an independent state; concurrent users cannot access each other's data
- API key security: Claude API keys are loaded from environment variables, never from the UI or source code
- Sample data transmission: only statistical aggregates and small samples ("d20 rows) are sent to the LLM — full datasets are never transmitted
- HTTPS-only deployment recommended; reverse proxy config provided for Nginx
- Optional PII scrubber: regex-based module masks emails, phone numbers, and common ID patterns before LLM transmission

12. Performance Benchmarks

Test Environment

AWS EC2 t3.medium (2 vCPU, 4 GB RAM), Python 3.11, Streamlit 1.32, Claude 3 Haiku (default).

Results

- 1K rows, 10 columns !' full analysis in 3.2 seconds
- 50K rows, 25 columns !' full analysis in 8.7 seconds
- 500K rows, 15 columns !' profiling 12s, LLM 4s, charts 6s !' 22s total
- Concurrent users: tested up to 10 simultaneous sessions with < 5% performance degradation
- API cost (Claude Haiku): ~\$0.002 per analysis (average 1,800 input tokens)

13. User Experience Design

The UI follows a progressive disclosure pattern to avoid overwhelming non-technical users.

- Step 1: Upload file (drag & drop or file picker)
- Step 2: Dataset preview with type annotations (auto)
- Step 3: "Analyse" button triggers pipeline
- Step 4: Progress bar with stage labels (Profiling !' Analysing !' Rendering)
- Step 5: Tabbed results view: Summary | Charts | Anomalies | Raw Profile
- Step 6: Export options (PDF, HTML, JSON)

Accessibility: WCAG 2.1 AA compliant colour contrast. Keyboard-navigable. Responsive layout (works on tablets for field use).

14. Deployment & Infrastructure

Local Development

- `pip install -r requirements.txt`
- `export ANTHROPIC_API_KEY=sk-...`
- `streamlit run app.py`

Docker

A multi-stage Dockerfile is provided. The production image is 280 MB (Alpine base). Docker Compose config includes: app, nginx reverse proxy, and optional Redis for session caching.

AWS Deployment

- EC2 t3.medium with Elastic IP
- Application Load Balancer with HTTPS termination (ACM certificate)
- CloudWatch alarms for CPU > 70% and error rate > 1%
- Auto Scaling Group: min 1, max 4 instances

15. Testing & Quality Assurance

Unit Tests (pytest)

- Data profiler: 47 test cases covering edge cases (empty columns, all-null, mixed types)
- Prompt builder: snapshot tests for 12 dataset archetypes
- Response parser: validation of Claude's JSON output structure

Integration Tests

- End-to-end pipeline tests using 8 sample datasets (retail, finance, healthcare, IoT)
- Mock Claude API responses for CI/CD (no API cost in automated testing)
- Load testing: Locust scripts simulating 50 concurrent users

Code Quality

- Type hints throughout (mypy strict mode)
- Black formatter + isort
- Ruff linter (0 warnings policy)
- Pre-commit hooks enforced

16. Known Limitations & Mitigations

- No real-time streaming data support (planned in v2.0 via Kafka connector)
- Claude API rate limits may affect high-concurrency deployments !' mitigated by request queuing
- Very wide datasets (> 200 columns) require column selection to fit context window
- Free-text columns are summarised by word frequency, not semantic analysis (semantic analysis in roadmap)
- No native database connector (CSV/Excel only) — JDBC/ODBC connector in roadmap

17. Roadmap & Future Enhancements

v1.1 (Q3 2026)

- PostgreSQL & MySQL direct connection
- Scheduled analysis with email digest
- Slack/Teams webhook integration for anomaly alerts

v2.0 (Q4 2026)

- Real-time streaming via Kafka
- Multi-dataset join and comparison
- Role-based access control for team deployments
- Self-hosted LLM support (Ollama)

18. Business Impact & ROI

Based on a pilot with 3 internal teams (n=14 analysts):

- Average analysis time reduced from 2.4 hours to 55 minutes (61% reduction)
- Ad-hoc report requests to the data team reduced by 38% in 4 weeks
- Non-technical stakeholders completed 85% of standard analyses independently
- Estimated annual analyst time saved: 312 hours per 5-person team
- Tool cost (API + compute): ~\$80/month for a 10-person team at 200 analyses/month

19. Conclusion

AI Data Insights Engine demonstrates that LLM-augmented analytics tools can deliver significant productivity gains without requiring infrastructure overhaul or retraining of non-technical users. By combining the interpretive power of Claude with the robustness of the Python data science ecosystem, the project achieves a rare balance of depth and accessibility.

The open-source release invites the community to extend the tool with new connectors, visualisation types, and LLM backends. The author welcomes contributions and enterprise collaboration inquiries.

Rahul Sangral

rahul.rishusangral@gmail.com · India

20. References & Appendix

References

- Anthropic (2024). Claude 3 Model Card. anthropic.com
- IDC (2024). Data Analyst Productivity Report. idc.com
- Streamlit (2024). Documentation. docs.streamlit.io
- Pandas Development Team (2024). Pandas 2.1 Release Notes.
- NIST (2023). Privacy Framework for AI Systems.

Appendix A — Sample Prompt Template

You are a senior data analyst reviewing a business dataset. Below is a structured profile of the dataset. Provide a JSON response with keys: "summary" (3-paragraph executive overview), "trends" (array of top 5 trend observations), "anomalies" (array of flagged rows/values with explanation), "recommended_charts" (array of chart specs with column names and chart type), "follow_up_questions" (array of 3 questions the analyst should investigate next).

Appendix B — Configuration Reference

- MAX_FILE_SIZE_MB: 500
- DEFAULT_MODEL: claude-3-haiku-20240307
- FALLBACK_MODEL: claude-3-sonnet-20240229
- ANOMALY_ZSCORE_THRESHOLD: 3.0
- MAX_SAMPLE_ROWS: 20
- CHUNK_SIZE_ROWS: 50000